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Saveetha Institute of Medical And Technical Sciences
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Integrated Hotel Booking And Guest Experience Portal

CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA4305-INTERNET PROGRAMMING

to the award of the degree of

BACHELOR OF ENGINEERING

IN

**COMPUTER SCIENCE AND ENGINEERING
(CSE)**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Integrated Hotel Booking And Guest Experience Management System” has been carried out by P.Harika (192311243) under the supervision of DR D Sudhagar and is submitted in partial fulfilment of the requirements for the current semester of the B.E Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, P.Harika(192311243) of the CSE , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Integrated Hotel Booking And Guest Experience Management System” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

In accordance with capstone evaluation guidelines, the detailed individual responsibilities and specific architectural contributions of each group member are delineated below:

Student Name / Reg No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. (%)
P.Harika (192311243)	User Portal & User Authentication	Designed and developed user registration, login, authentication, session management, hotel browsing, room viewing	Tested registration forms, login validation, session handling, hotel browsing, room selection, booking functionality, and invalid-input handling	Abstract, Chapter 1, requirements documentation, and User Portal documentation	33.3%
K.Mrudhula (192372290)	Admin Panel – Hotel & Booking Control	Developed administrator login, hotel management, room management, customer management, booking management, cancellation management,	Tested admin authentication, hotel and room operations, booking management, cancellation functionality, availability validation, and unauthorized access	Chapter 2, Admin Panel documentation, existing-system analysis, and related documentation	33.3%
N. Vishnu Sai Sri (192372039)	Payment & Billing and System Integration	Developed payment processing, payable amount calculation, payment status management, invoice generation, email notification	Conducted payment testing, invoice verification, email notification testing, database integration testing, debugging, and system-level testing	Chapters 3, 4, 5, testing documentation, and Appendices	33.4%
Total	Complete Integrated Hotel Booking and Guest Experience Management System	Full End-to-End Web Application Development	Functional, Integration & System Testing	Complete Technical Documentation	100%

ABSTRACT

The Integrated Hotel Booking and Guest Experience Management System is a web-based application designed to provide a centralized platform for managing hotel information, room availability, reservations, guest details, and payment-related activities. In traditional hotel booking processes, reservations are often handled through phone calls, manual records, and other time-consuming procedures. Guests may also face difficulties in checking room availability and obtaining booking information in real time. The proposed system addresses these challenges by integrating hotel browsing, room booking, guest management, payment processing, and booking information within a single web-based platform.

The system allows users to register and log in, browse cities and hotels, view hotel and room details, check room availability, make reservations, complete payment-related activities, and view their bookings. The system also maintains customer and booking information in a structured database. The payment and billing module processes booking information, calculates the final payable amount, validates payment details, generates billing information, and maintains payment status. Email notifications are provided to confirm successful booking and payment transactions. On the administrative side, authorized administrators can manage hotels, rooms, customers, bookings, cancellations, and reports, providing better control over hotel operations.

The application demonstrates the practical implementation of Internet Programming concepts, including HTML5, CSS3, Bootstrap 5, Core PHP, MySQL, sessions, SQL queries, and server-side validation. PHP is used to implement business logic and handle hotel, room, booking, and payment operations, while MySQL provides structured storage for customer, hotel, room, booking, and payment information. Session-based authentication maintains secure user access across different pages, while SQL queries and JOIN operations support efficient retrieval of hotel, room, and booking information. The system is developed and tested using a web-based environment with XAMPP, Apache, MySQL, and browser-based testing.

The proposed system provides a systematic and efficient approach to hotel reservation and guest experience management by combining hotel discovery, room availability, booking, payment, billing, notification, and administrative management in one platform. It can reduce manual work, minimize booking errors and double bookings, improve the accuracy of hotel records, and provide guests with convenient access to reservation services from anywhere. The system also improves operational efficiency by providing administrators with centralized control over hotel and booking information. Future enhancements can include machine-learning-based dynamic room pricing, mobile application support, loyalty and personalization features, personalized offers, targeted notifications, and automated chatbots for booking assistance, frequently asked questions, and cancellations.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbr.	Description	Standard / Unit
IP	Internet Programming	Course / Domain
HTML	HyperText Markup Language	Web Standard
CSS	Cascading Style Sheets	Web Standard
JS	JavaScript	Programming Language
DOM	Document Object Model	Web Programming
UI	User Interface	Software
UX	User Experience	Software Design
HTTP	HyperText Transfer Protocol	Web Protocol
HTTPS	HyperText Transfer Protocol Secure	Secure Web Protocol
URL	Uniform Resource Locator	Web Address
API	Application Programming Interface	Software Interface
JSON	JavaScript Object Notation	Data Format
SQL	Structured Query Language	Database Language
DB	Database	Data Storage
DBMS	Database Management System	Software
CRUD	Create, Read, Update, Delete	Database Operations
REST	Representational State Transfer	Web Architecture
AJAX	Asynchronous JavaScript and XML	Web Technology
PHP	Hypertext Preprocessor	Server-Side Language
IDE	Integrated Development Environment	Software Tool
IDE	Integrated Development Environment	Development Tool
MVC	Model-View-Controller	Software Architecture
JWT	JSON Web Token	Authentication / Security
Auth	Authentication	Security
RBAC	Role-Based Access Control	Security
OTP	One-Time Password	Authentication
ERD	Entity-Relationship Diagram	Database Design
LMS	Learning Management System	Educational System
SDG	Sustainable Development Goals	UN Framework
OWASP	Open Worldwide Application Security Project	Web Security
WCAG	Web Content Accessibility Guidelines	Accessibility Standard
FR	Functional Requirement	Requirement Specification
NFR	Non-Functional Requirement	Requirement Specification
API	Application Programming Interface	Software

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

In the present digital and competitive hospitality industry, hotels require efficient systems to manage room information, customer details, reservations, payments, and guest services. Traditional hotel booking processes often involve phone calls, manual records, paperwork, and direct communication between guests and hotel staff. These methods can be time-consuming and may make it difficult for guests to obtain accurate information about room availability, pricing, and booking status.

Guests increasingly expect convenient online services through which they can browse hotels, view available rooms, check room details, make reservations, and receive booking confirmations without depending completely on hotel staff. At the same time, hotel administrators require a centralized system to manage hotels, rooms, customers, bookings, cancellations, and payment information efficiently.

The Integrated Hotel Booking and Guest Experience Management System is proposed as a web-based application to provide a centralized platform for hotel reservation and guest management. The system allows users to browse cities and hotels, view room details, check availability, make bookings, complete payment-related activities, and view their booking information. The system also provides administrative facilities for managing hotels, rooms, users, bookings, and reports.

The proposed system applies Internet Programming concepts to develop an interactive and accessible hotel reservation platform. HTML5, CSS3, and Bootstrap 5 can be used to develop the responsive user interface, while Core PHP is used for server-side business logic and MySQL is used for structured data storage. Sessions, SQL queries, validation, payment processing, invoice generation, and email notifications support the major operations of the system.

1.2 Need for the Project

Traditional hotel booking methods require guests to depend on phone calls, physical visits, or manual communication to obtain information about hotels and room availability. Such processes can increase the time required to complete reservations and may create difficulties in maintaining accurate booking information.

Manual record-keeping can also result in outdated booking information, data inconsistencies, and the possibility of double bookings. Hotel staff may need to maintain separate records for customers, rooms, hotels, and reservations, making it difficult to efficiently retrieve and update information.

Guests also require a self-service platform where they can browse hotels, select rooms, make reservations, complete payment procedures, and view their booking details conveniently. Therefore, there is a need for an integrated web-based system that combines hotel management,

room availability, booking, customer management, payment, billing, and guest services within a single platform.

The Integrated Hotel Booking and Guest Experience Management System addresses this need by providing:

- Centralized hotel and room information management.
- Online hotel and room browsing.
- Real-time room availability checking.
- Online room reservation functionality.
- Customer and booking information management.
- Payment and billing management.
- Invoice and booking confirmation generation.
- Email notification for successful transactions.
- Administrative management of hotels, rooms, users, and bookings.
- Reduced manual effort and improved booking accuracy.

The need for centralized management is particularly important because the project identifies problems such as the lack of real-time room availability, manual record keeping, double-booking risks, and data inconsistency in traditional processes.

1.3 Problem Statement

The engineering problem addressed by this project is to design and implement a web-based hotel booking and guest experience management system capable of managing hotel information, room availability, customer details, reservations, payments, billing, and booking notifications.

The system must allow users to register and securely log in, browse cities and hotels, view hotel and room information, check room availability, make reservations, and access their booking details. The system must also process payment information, calculate the payable amount, maintain payment status, and generate appropriate billing information.

From the administrative perspective, the system should provide facilities to manage hotels, rooms, customers, bookings, cancellations, and reports. The system should reduce the difficulties associated with manual hotel booking and record maintenance while minimizing the possibility of duplicate bookings and inconsistent data.

Therefore, the proposed system aims to provide an integrated and user-friendly platform that improves booking efficiency, data organization, guest convenience, and hotel operational management.

1.4 Project Aim

The primary aim of this project is to develop an **Integrated Hotel Booking and Guest Experience Management System**, a web-based application that combines hotel browsing, room availability, online reservation, customer management, payment processing, billing, and administrative management into a single platform.

The system aims to provide guests with a convenient method of discovering hotels, selecting rooms, making reservations, completing payment-related activities, and viewing booking information. It also aims to provide hotel administrators with centralized facilities for managing hotels, rooms, customers, bookings, cancellations, and reports.

1.5 Project Objectives

The project is developed with the following objectives:

- **Objective 1:** To develop a web-based platform for integrated hotel booking and guest experience management.
- **Objective 2:** To provide secure user and administrator registration/login and session-based authentication.
- **Objective 3:** To allow users to browse cities, hotels, rooms, and room details dynamically.
- **Objective 4:** To provide room availability checking before confirming a reservation.
- **Objective 5:** To implement an online hotel and room booking facility for guests.
- **Objective 6:** To maintain customer, hotel, room, and booking information using a structured MySQL database.
- **Objective 7:** To implement payment processing and calculate the final payable booking amount.
- **Objective 8:** To generate billing information and maintain payment status for completed transactions.
- **Objective 9:** To provide email notifications for booking and payment confirmation.
- **Objective 10:** To provide administrative functionality for managing hotels, rooms, users, bookings, cancellations, and reports.
- **Objective 11:** To reduce manual effort and minimize errors associated with traditional hotel booking and record management.
- **Objective 12:** To apply relevant Internet Programming technologies such as HTML5, CSS3, Bootstrap 5, Core PHP, MySQL, sessions, and SQL queries in a practical web application.

1.6 Scope

The scope of the **Integrated Hotel Booking and Guest Experience Management System** includes the development of a web-based application for managing hotel information, room reservations, guest information, payment and billing, and administrative activities.

Included Scope

- User registration and login.
- Administrator authentication.
- Session-based authentication.
- City and hotel browsing.
- Hotel details display.
- Room details and room gallery display.
- Room availability checking.
- Online room booking.
- Customer information management.
- Booking management.
- Booking cancellation.
- Payment processing.
- Payment validation.

The project specifically includes separate **Admin Panel**, **User Portal**, and **Payment & Billing System** modules.

Excluded Scope

The initial system does not focus on advanced hotel-industry functions such as complete hotel staff payroll management, restaurant inventory management, housekeeping automation, airline or travel package management, or large-scale external hotel-chain integration.

Advanced features such as machine-learning-based dynamic pricing, dedicated mobile applications, advanced loyalty programs, personalized offers, and automated chatbot-based guest assistance are considered possible future enhancements.

Assumptions

- Users provide valid registration and booking information.
- Hotel and room information stored in the database is accurate.
- Room availability information is updated through the system.

- Users have access to a compatible web browser and internet connection.
- The application server and MySQL database are available during system operation.
- Payment information supplied during the transaction is valid.
- Email notification services are available for sending confirmations.

Boundary Conditions

- Only registered/authenticated users can access personalized booking functions.
- Administrative functions are accessible only to authorized administrators.
- Room bookings depend on the availability of the selected room.
- Booking confirmation depends on successful validation and processing of booking information.
- Payment status is maintained according to the transaction information available in the system.
- Invoice generation depends on valid booking and payment information.
- User access is controlled using session-based authentication.
- Hotel and room information is limited to the data maintained within the system.

1.7 Expected Engineering Outcome

The expected outcome of the project is a functional web-based **Integrated Hotel Booking and Guest Experience Management System** that provides an organized workflow for hotel discovery, room reservation, payment, billing, and hotel administration.

The expected **user-side execution flow** is:

User Registration → Login → Browse Cities → Select Hotel → View Rooms → Check Availability → Book Room → Payment → Invoice/Booking Confirmation → Email Notification → View Bookings

The expected **admin-side execution flow** is:

Admin Login → Admin Dashboard → Manage Hotels → Manage Rooms → Manage Users → Manage Bookings → Manage Cancellations → View Reports

The expected **payment and billing flow** is:

Booking Details → Payment Validation → Amount Calculation → Payment Processing → Payment Status Update → Invoice Generation → Email Confirmation.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

To establish a suitable foundation for the Integrated Hotel Booking and Guest Experience Management System, a structured review of existing hotel reservation systems, web-based hotel booking applications, hotel management platforms, and technology-enabled guest service systems was considered. The review focused on systems that support hotel and room management, online reservations, customer information management, payment processing, billing, and guest experience enhancement.

The reviewed systems were evaluated across four major analytical dimensions: (1) hotel and room information management, (2) online booking and availability management, (3) payment, billing, and notification support, and (4) administrative management and guest usability. These dimensions were selected to identify the capabilities and limitations of existing approaches and to determine the requirements for an integrated web-based hotel reservation solution.

The project literature review includes studies related to online hotel reservations, web-based hotel booking applications, and modern technology-based hotel management solutions. The reviewed work includes survey-based guest satisfaction analysis, PHP-MySQL hotel booking systems, and MERN-based hotel booking applications.

2.2 Review of Existing Work

Traditional Manual Hotel Booking Systems

Traditional hotel reservations are commonly performed through phone calls, direct visits, paper-based records, and manual communication between guests and hotel staff. Hotel employees may maintain separate records for rooms, customers, and reservations.

Although manual methods can support basic booking operations, they require considerable human effort and may result in outdated information, data inconsistencies, and booking errors. The absence of a centralized system also makes it difficult for guests to check room availability and complete reservations independently.

Web-Based Hotel Booking Systems

Web-based hotel booking systems provide an online platform through which users can browse hotels, view available rooms, check room details, and make reservations. These systems reduce the dependency on manual booking procedures and provide users with access to reservation services through a web browser.

Existing PHP-MySQL-based hotel booking applications demonstrate the use of CRUD operations and session-based authentication for managing hotel information and reservations. Such systems can reduce manual work and provide a relatively simple and scalable approach to hotel reservation management.

Online Hotel Reservation and Guest Satisfaction Systems

Some existing research focuses on the relationship between online hotel reservation systems and guest satisfaction. These studies examine aspects such as user experience, interface design, accessibility, and the convenience provided by online reservation platforms.

Such approaches demonstrate that technology can contribute to improving the guest experience. However, survey-based approaches primarily focus on evaluating user satisfaction rather than implementing a complete integrated technical system for hotel, room, booking, payment, and administrative management.

Modern Hotel Booking Applications

Modern hotel booking applications use web technologies and frameworks to provide dynamic interfaces, online reservation functionality, database integration, and secure authentication. MERN-based applications, for example, can use REST APIs and JWT authentication to provide modern and scalable hotel booking services.

These systems provide advantages such as responsive user interfaces and scalability. However, their implementation can involve a more complex technology stack and may require higher technical expertise compared with simpler PHP-MySQL-based systems.

Technology-Based Guest Experience Management

Modern hospitality systems increasingly use technology to improve guest interaction and hotel service management. Technology-enabled guest experience systems can provide users with convenient access to hotel information and reservation services while helping hotel administrators manage operational information.

However, individual systems may focus on specific areas such as booking, guest satisfaction, or hotel management rather than combining hotel browsing, room availability, booking, payment, billing, notification, and administration in one integrated web application.

2.3 Comparative Analysis

Table 2.1 presents a comparative evaluation of existing approaches against the proposed **Integrated Hotel Booking and Guest Experience Management System**.

Table 2.1 Comparative Analysis of Existing and Proposed Systems

Platform System	Hotel & Room Management	Online Booking	Payment & Billing	Guest Admin / Management	Integrated Usability
Manual Records Phone Booking	Moderate (Manual entry)	Low (Manual process)	Low	Low	Low
Traditional Hotel	High	Moderate	Moderate	High	Moderate

Management Systems					
Web-Based PHP-MySQL Booking Systems	High	High	Moderate	High	High
MERN-Based Hotel Booking Applications	High	High	Moderate	High	High
Guest Satisfaction UX-Based Systems	Low	Moderate	Low	Moderate	Moderate
Integrated Hotel Booking and Guest Experience Management System (Proposed)	High (Centralized hotel and room management)	High (Online availability and reservation)	High (Payment, billing and invoice)	High (User and admin portals)	High (Integrated web platform)

The proposed system provides an integrated combination of hotel and room management, online reservation, customer management, payment and billing, and administrative functionality. The project specifically separates these functions into the **Admin Panel, User Portal, and Payment & Billing System** modules.

2.4 Research / Engineering Gap

The review of existing approaches identifies several important engineering gaps:

- **Fragmented Hotel Information:** Existing manual and basic reservation approaches may maintain hotel, room, customer, and booking information separately. This makes centralized information management difficult.
- **Lack of Real-Time Availability:** Traditional booking methods do not provide guests with convenient access to current room availability, which can increase the possibility of booking conflicts.
- **Manual Booking and Record Maintenance:** Phone-based and paper-based booking processes require significant manual effort and can result in outdated or inconsistent booking records.
- **Risk of Double Booking:** When room status and reservation records are maintained manually, the possibility of duplicate or conflicting reservations can increase.

- **Limited Self-Service Booking:** Traditional approaches do not provide guests with a complete self-service platform for browsing hotels, selecting rooms, making bookings, and viewing reservations.
- **Separation of Booking and Payment Processes:** Some systems may focus mainly on reservations without providing integrated payment validation, billing information, invoice generation, and payment-status management.
- **Limited Guest Notification:** Manual reservation approaches may depend on direct communication for booking confirmation, whereas an integrated system can provide email notifications after successful booking and payment.
- **Lack of Unified Administrative Control:** Hotel administrators require a centralized interface for managing hotels, rooms, customers, bookings, cancellations, and reports.
- **Need for an Integrated Web-Based Solution:** Existing approaches often concentrate on individual functions such as reservation, guest satisfaction, or application development. There is a need for a unified web application that combines these functions into a single workflow.

The identified problem areas are consistent with the project requirements, which highlight manual booking, lack of centralized real-time availability, double-booking risks, data inconsistency, and the need for a secure and user-friendly online reservation portal.

2.5 Proposed Contribution

The proposed **Integrated Hotel Booking and Guest Experience Management System** addresses the identified gaps through an integrated web-based approach:

- **Unified Hotel and Room Management:** Provides centralized management of hotel, room, pricing, and availability information.
- **Online Hotel Discovery:** Allows users to browse cities, hotels, room details, and available accommodation through a web interface.
- **Integrated Room Availability Checking:** Checks room availability before confirming a booking, helping reduce the possibility of double booking.
- **Online Reservation:** Allows authenticated users to select rooms and submit booking information without depending on manual hotel communication.
- **Centralized Customer and Booking Management:** Maintains customer and booking information in a structured MySQL database for efficient retrieval and management.
- **Integrated Payment and Billing:** Processes payment-related information, calculates the final payable amount, updates payment status, and maintains payment records.
- **Invoice Generation:** Generates a billing summary containing customer details, hotel and room information, check-in and check-out dates, number of nights, total amount, and payment status.

- **Email Notification:** Provides booking and payment confirmation through email after successful transactions, improving communication with guests.
- **Administrative Management:** Provides authorized administrators with facilities to manage hotels, rooms, users, bookings, cancellations, and reports.
- **Session-Based Authentication:** Maintains authenticated user sessions across pages and restricts unauthorized access to protected functions.
- **Responsive Web Interface:** Uses HTML5, CSS3, and Bootstrap 5 to provide a responsive and user-friendly interface for hotel and room browsing and booking activities.

The overall contribution of the proposed system is the integration of **hotel discovery, room management, availability checking, reservation, payment, billing, notification, and administrative control** within a single web-based application. This provides a systematic approach to hotel reservation management while improving guest convenience and reducing manual operational effort.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

To establish clear engineering benchmarks, the system requirements for the **Integrated Hotel Booking and Guest Experience Management System** were identified based on the needs of guests, hotel administrators, and the overall hospitality environment. These requirements were translated into functional and non-functional specifications covering user management, hotel and room management, room availability, online booking, payment and billing, notifications, and administrative management.

Stakeholder Requirements

Stakeholder	Primary Need / Goal	Derived Engineering Requirement
Guest / User	Search and book suitable hotel rooms conveniently	Online hotel browsing, room details, availability checking, booking, payment, and booking history
Hotel Administrator	Manage hotels, rooms, customers, and reservations	Centralized hotel, room, user, booking, cancellation, and report management
Payment System	Process and maintain booking payment information	Payment validation, amount calculation, payment-status management, and invoice generation
Hotel Management	Maintain accurate and organized reservation information	Reliable database storage and retrieval of hotel, room, customer, and booking records
System Administrator	Maintain secure and reliable system operation	Authentication, authorization, database management, validation, and controlled access

Functional and Non-Functional Requirements

Requirement ID	Category	Description	Measurable Target
FR-01	Functional	User registration and login	Successful authentication for registered users
FR-02	Functional	Administrator authentication	Only authorized administrators can access admin functions

FR-03	Functional	Hotel management	Admin can create, read, update, and delete hotel information
FR-04	Functional	Room management	Admin can create, read, update, and delete room information
FR-05	Functional	Hotel and room browsing	Users can browse cities, hotels, and available rooms
FR-06	Functional	Room availability checking	System checks room availability before booking confirmation
FR-07	Functional	Online room booking	Authenticated users can submit and confirm room reservations
FR-08	Functional	Booking management	Users can view their bookings and administrators can manage reservations
FR-09	Functional	Payment processing	System processes booking payment information and updates payment status
FR-10	Functional	Invoice generation	System generates billing information for completed transactions
FR-11	Functional	Email notification	Booking and payment confirmation is sent to the guest
FR-12	Functional	Customer management	Administrator can manage customer information
FR-13	Functional	Cancellation management	Administrator can manage booking cancellation records
FR-14	Functional	Reports	Administrator can access relevant booking and management information
NFR-01	Non-Functional	Usability	Provide a simple and responsive web interface
NFR-02	Non-Functional	Performance	Common browsing and booking operations should respond without noticeable delay
NFR-03	Non-Functional	Reliability	Maintain consistent hotel, room, booking, and payment records

NFR-04	Non-Functional	Security	Protect credentials and restrict unauthorized access
NFR-05	Non-Functional	Maintainability	Use modular PHP and database structures for easier maintenance
NFR-06	Non-Functional	Compatibility	System should operate through commonly used web browsers

The project implements separate **Admin Panel, User Portal, and Payment & Billing System** modules to satisfy these requirements.

3.2 Constraints & Applicable Standards

The development of the **Integrated Hotel Booking and Guest Experience Management System** is subject to technological, usability, security, performance, and resource constraints. Since the application is developed as an Internet Programming project, standard web-development practices are considered to improve reliability, accessibility, maintainability, and security.

Standard Guideline	Issuing Body	Scope & Requirements	Implementation in Proposed System
HTML5	W3C	Structured web-page development	Semantic structure for hotel, room, booking, and administrative pages
CSS3	W3C	Web presentation and responsive styling	Layout, styling, and responsive interface
Bootstrap 5	Open-source framework	Responsive web interface	Responsive cards, forms, navigation, and page components
PHP	PHP Community	Server-side web programming	Booking, hotel, room, payment, and business logic
MySQL	Oracle	Relational database management	Storage of users, hotels, rooms, bookings, and payment information
HTTP/HTTPS	IETF	Web communication	Communication between browser and application server
OWASP Web Security Guidelines	OWASP	Secure web application development	Authentication, input validation, session control, and secure database practices

The project specifically uses **Core PHP, MySQL, HTML5, CSS3, Bootstrap 5, SQL JOIN queries, sessions, XAMPP, Apache, and browser-based testing.**

batch_3

Major Constraints

- The system depends on the availability of a web browser and server environment.
- Room availability depends on the accuracy of database records.
- Booking confirmation depends on successful room availability validation.
- Payment processing depends on the payment information available to the system.
- Email confirmation depends on the availability of the configured email service.
- The initial system does not implement advanced AI-based dynamic pricing.
- The system is primarily designed for hotel booking and guest experience management.
- Database availability is required for persistent storage of hotel, room, customer, booking, and payment records.

3.3 Design Specifications

Table 3.3 defines the major design parameters considered for the proposed system.

Design Parameter	Specification Target	Engineering Unit	Priority
User registration	Complete registration with required details	User account	Critical
Authentication	Valid username/email and password	Login request	Critical
Hotel information	Accurate hotel details	Hotel records	High
Room information	Room type, price, and availability	Room records	High
Room availability	Check availability before confirmation	Room status	Critical
Booking	Complete reservation with required details	Booking record	Critical
Customer information	Store valid guest information	Customer record	High
Payment processing	Calculate and process booking amount	Currency / transaction	Critical
Payment status	Maintain current payment status	Payment record	Critical

Invoice generation	Generate complete billing summary	Invoice / booking	High
Email notification	Send booking/payment confirmation	Email notification	High
Booking history	Display previous and current bookings	Booking records	High
Database integrity	Consistent storage and retrieval	Database records	Critical
Interface responsiveness	Usable on common screen sizes	Web interface	Medium
Access control	User and administrator roles	User role	Critical
Validation	Prevent invalid or incomplete data	Form/request	High

The payment module specifically handles booking information, payable amount calculation, payment-status updates, invoice generation, and confirmation notifications.

3.4 Alternative Solutions & Evaluation

During system design, three alternative approaches were considered for implementing the **Integrated Hotel Booking and Guest Experience Management System**.

Alternative 1: Manual / Phone-Based Booking System

This approach stores hotel, room, customer, and booking information using paper records, spreadsheets, phone calls, or manually maintained documents. It is inexpensive and simple to start with, but it provides limited automation, weak data consistency, and no convenient real-time online booking workflow.

Alternative 2: Separate Web Applications

In this approach, different applications are used for hotel information, room booking, payment, customer management, and administration. Each individual application can perform its specific task, but users and administrators must work across multiple systems, resulting in distributed information and additional integration requirements.

Alternative 3: Integrated Web-Based Hotel Booking System

The proposed approach combines authentication, hotel management, room management, availability checking, online booking, customer management, payment processing, billing, email notification, and administration into one application. This provides centralized data management and a consistent user experience while remaining suitable for an Internet Programming project.

Evaluation of Alternatives

Evaluation Criterion	Weight	Manual / Phone-Based	Separate Web Applications	Integrated Proposed System
Hotel & Room Management	20%	2 / 5	4 / 5	5 / 5
Online Booking	25%	1 / 5	4 / 5	5 / 5
Payment & Billing	20%	2 / 5	4 / 5	5 / 5
Usability	15%	2 / 5	3 / 5	5 / 5
Maintenance & Integration	20%	3 / 5	2 / 5	4 / 5

Justification: The integrated web-based approach provides the best balance between functionality, usability, maintainability, and implementation feasibility. It eliminates the fragmentation associated with manual records and separate applications while providing a centralized platform for hotel booking and guest management.

3.5 Selected Design & Detailed Engineering Architecture

The proposed system is organized into **three major functional modules** operating as an integrated web-based pipeline.

Module 1: Admin Panel – Hotel and Booking Control

This module manages the administrative operations of the hotel booking system. The administrator can manage hotels, rooms, users, bookings, cancellations, and reports. Administrative authentication ensures that only authorized users can access management functions.

Module 2: User Portal – Booking & Guest Experience

This module provides the main interface for hotel guests. Users can register and log in, browse cities, view hotels and rooms, check room availability, make bookings, complete payment activities, and view their bookings.

The user portal dynamically retrieves hotel, room, customer, and booking information from the MySQL database.

batch_3

Module 3: Payment & Billing System

This module manages payment-related booking operations. It processes payment requests, calculates the final payable amount, validates booking information, updates payment status, maintains payment records, and generates billing information.

The invoice contains relevant information such as customer name, hotel and room details, check-in/check-out dates, number of nights, total amount, and payment status. Email notification is then used to confirm successful booking and payment transactions.

Overall System Flow

User Registration → Login → Browse Cities → Select Hotel → View Rooms → Check Availability → Book Room → Payment → Invoice Generation → Email Confirmation → View Bookings

Admin Flow

Admin Login → Admin Dashboard → Manage Hotels → Manage Rooms → Manage Users → Manage Bookings → Manage Cancellations → Reports

3.6 Trade-off Analysis

Engineering design requires selecting suitable technologies and approaches while considering simplicity, performance, maintainability, security, and functionality.

Design Decision	Alternative Option	Evaluated Advantage	Evaluated Penalty	Resolution & Justification
Integrated Web Application	Separate applications	Centralized information and simpler workflow	Requires integration of multiple modules	Selected integrated architecture for unified hotel management
Core PHP	MERN / other frameworks	Simple deployment and suitable for academic web development	Less feature-rich than some modern frameworks	Selected because it is suitable for the project's implementation requirements
Relational Database	NoSQL Database	Strong relationships between hotels, rooms, users, bookings, and payments	Requires predefined schema	Selected MySQL because hotel booking data is highly relational
Session-Based Authentication	Token-based authentication	Simple and suitable for server-side PHP applications	Requires session management	Selected for maintaining logged-in user state
Availability Validation	Manual room-status checking	Automated checking reduces booking conflicts	Requires accurate database state	Selected to reduce double-booking risk

Server-Side Validation	Client-only validation	Provides stronger control over submitted data	Requires additional backend processing	Selected as the primary validation mechanism
Responsive Web Interface	Fixed-size interface	Works across different screen sizes	Requires additional responsive design	Selected for improved guest accessibility
SQL JOIN Queries	Separate database queries	Efficient retrieval of related hotel and room information	Query complexity can increase	Selected for hotel-room and booking relationships

The project uses SQL JOIN queries for hotel-room mapping and sessions to maintain user login across pages such as Home, Book Room, and My Bookings.

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3.7 Sustainability, Ethics, Safety, Risk & Security

The **Integrated Hotel Booking and Guest Experience Management System** considers sustainability, ethical use of customer information, system safety, operational risk, and security throughout its design.

Domain	Consideration	Evidence Evaluated	Impact on System Design
Sustainability	Reduction of paper-based booking records	Digital reservation and database storage reduce dependency on physical records	Hotel, customer, room, and booking information are maintained electronically
Ethics	Responsible use of customer information	Customer and booking information are sensitive business/user records	Access is restricted to authorized users
Fairness	Consistent booking processing	Manual booking can result in inconsistent handling	Predefined booking and availability rules are applied
Safety	Prevention of incorrect booking operations	Invalid inputs may create incorrect or inconsistent records	Input validation and controlled booking operations are applied

Security	Protection against unauthorized access	User credentials and booking/payment records require protection	Authentication, authorization, session control, and validation are used
Data Integrity	Accurate reservation records	Incorrect or duplicate records can affect hotel operations	Database validation and controlled CRUD operations are applied

The system applies authentication, sessions, backend validation, database management, and controlled administrative access as part of its implementation.

Risk Register and Mitigation Strategy

Identified Risk	Likelihood	Severity	Risk Level	Mitigation Strategy	Residual Risk
Unauthorized account access	Medium	High	High	Authentication and session-based access control	Low
Double booking	Medium	High	High	Check room availability before confirming booking	Low
Incorrect room information	Medium	Medium	Medium	Validate and review hotel/room records	Low
Incorrect booking information	Medium	Medium	Medium	Form validation and booking verification	Low
Invalid payment information	Medium	High	High	Payment validation before processing	Low
Duplicate payment records	Low	High	Medium	Validate booking and payment status before recording transaction	Low
Database failure / data loss	Low	High	Medium	Regular database backup and validation	Low
Invalid user input	High	Medium	High	Client- and server-side validation	Low
Unauthorized modification of records	Low	High	Medium	Administrator authentication and access restrictions	Low

Email notification failure	Medium	Medium	Medium	Error handling and confirmation-status checking	Low
Server unavailability	Medium	Medium	Medium	Proper server configuration and error handling	Low
Incorrect cancellation handling	Low	Medium	Low	Controlled cancellation operations and record validation	Low

Security Considerations

The system incorporates several security measures to protect hotel and customer information:

- User authentication before accessing personalized functions.
- Administrator authentication for administrative operations.
- Session-based access control.
- Input validation for forms and booking information.
- Controlled database operations.
- Restriction of administrative functions to authorized users.
- Validation of booking information before payment processing.
- Protection of customer and booking records from unauthorized modification.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The **Integrated Hotel Booking and Guest Experience Management System** was developed using an iterative software development approach, where the system was divided into smaller functional modules. Each module was designed, implemented, tested, and integrated progressively. This approach helped in identifying errors at an early stage and ensured that the user portal, admin panel, booking, payment, billing, and notification components worked together correctly.

The development process focused on separating the user interface, application logic, and database operations. This separation improves maintainability and allows individual modules to be modified or tested without affecting the complete system. The major development activities included hotel and room management, user authentication, online booking, payment processing, invoice generation, email notification, and administrative management.

Resource Category	Resource Specification	Purpose & Architectural Role
Development Hardware	Computer/Laptop with adequate RAM and storage	Application development, testing, and execution
Operating System	Windows / Linux	Development and execution environment
Web Browser	Google Chrome / Microsoft Edge	Running and testing the web application
Database	MySQL relational database	Storage of users, hotels, rooms, customers, bookings, and payment records
Testing Data	Sample hotels, rooms, customer accounts, and booking details	Functional and system testing
Version Control	Git / GitHub	Source-code management and project submission
Development Server	XAMPP with Apache and MySQL	Local application execution and database testing

4.2 Modern Software Tools and Frameworks

Table 4.2 presents the major technologies and tools used for developing the proposed system.

Tool / Technology	Type / Layer	Functional Scope	Project Phase
HTML5	Frontend	Structure of hotel, room, booking, and administrative web pages	Development
CSS3	Frontend	Styling, layout, and responsive design	Development
Bootstrap 5	Frontend Framework	Responsive cards, forms, navigation, and interface components	Development
JavaScript	Frontend Logic	Dynamic interaction and client-side validation	Development
Core PHP	Backend	Server-side business logic and request processing	Development
MySQL	Database	Storage of users, hotels, rooms, bookings, customers, and payments	Data Management
SQL	Database Query Language	Data retrieval, insertion, updating, deletion, and JOIN operations	Development
Sessions	Authentication	Maintaining authenticated user state across pages	Development
XAMPP	Development Server	Apache server and MySQL environment	Development / Testing
Git	Version Control	Source-code version management	Development
GitHub	Repository	Code storage and project collaboration/submission	Submission
Web Browser	Testing	Application execution and interface testing	Verification
VS Code	Development Environment	Source-code development and debugging	Development

The project uses HTML5, CSS3, Bootstrap 5, Core PHP, MySQL, SQL JOIN queries, XAMPP, Apache, and browser-based testing as its main implementation technologies.

4.3 Test Plan & Verification

A comprehensive testing process was performed to verify the functionality and reliability of the system. **Unit testing** was used to validate individual modules, **integration testing** was used to verify communication between modules, and **system testing** was performed to evaluate the complete hotel booking workflow.

Test Module	Test Method & Procedure	Acceptance Criterion	Outcome
User Registration	Enter valid customer details and submit registration form	User account should be created successfully	Pass
User Login	Enter valid and invalid credentials	Valid user should log in; invalid user should be rejected	Pass
Admin Login	Enter administrator credentials	Authorized administrator should access admin dashboard	Pass
Hotel Management	Add, edit, and delete hotel information	Hotel records should be updated correctly	Pass
Room Management	Add, edit, and delete room information	Room records should be updated correctly	Pass
Hotel Browsing	Select city and browse hotels	Relevant hotels should be displayed	Pass
Room Details	Select a hotel and view rooms	Correct room details should be displayed	Pass
Room Availability	Select booking dates and room	Available room should be identified correctly	Pass
Room Booking	Enter valid booking information and submit	Booking should be created successfully	Pass
Booking Management	View booking information	Correct booking details should be displayed	Pass
Payment Processing	Submit valid payment information	Payment should be processed and status updated	Pass
Invoice Generation	Complete successful booking/payment	Billing summary should be generated correctly	Pass
Email Notification	Complete successful transaction	Booking/payment confirmation should be sent	Pass
Booking Cancellation	Cancel a valid booking	Cancellation should be recorded correctly	Pass
Customer Management	Admin views and manages customer records	Customer information should be maintained correctly	Pass

Input Validation	Enter incomplete/invalid data	Appropriate validation message should be displayed	Pass
Unauthorized Access	Attempt to access protected admin pages without authorization	Access should be denied	Pass

Requirement Verification and Traceability Matrix

Req ID	Specification Description	Observed Result	Compliance Status
FR-01	User Registration and Login	Registration and authentication completed successfully	Fully Compliant
FR-02	Administrator Authentication	Admin access restricted to authorized users	Fully Compliant
FR-03	Hotel Management	Hotel records successfully created and updated	Fully Compliant
FR-04	Room Management	Room records successfully maintained	Fully Compliant
FR-05	Hotel and Room Browsing	Hotels and rooms displayed according to selected information	Fully Compliant
FR-06	Room Availability	Availability checked before booking confirmation	Fully Compliant
FR-07	Online Room Booking	Booking successfully created and stored	Fully Compliant
FR-08	Booking Management	Booking information successfully displayed and managed	Fully Compliant
FR-09	Payment Processing	Payment information processed and status updated	Fully Compliant
FR-10	Invoice Generation	Billing summary generated successfully	Fully Compliant
FR-11	Email Notification	Confirmation notification generated after transaction	Fully Compliant
FR-12	Customer Management	Customer information successfully maintained	Fully Compliant

FR-13	Cancellation Management	Booking cancellation operation completed successfully	Fully Compliant
FR-14	Reports	Administrative information available for review	Fully Compliant
NFR-01	Usability	Interface tested through normal user workflow	Fully Compliant
NFR-02	Performance	Major operations completed within acceptable response time	Fully Compliant
NFR-03	Reliability	Hotel, room, booking, and payment records maintained correctly	Fully Compliant
NFR-04	Security	Authentication and access restrictions verified	Fully Compliant

4.4 Results & Experimental Evaluation

The completed system was evaluated using representative hotel booking scenarios covering different stages of the user and administrator workflow. The testing focused on whether users could successfully register, log in, browse cities and hotels, view rooms, check availability, make bookings, complete payment-related activities, receive booking confirmation, and view their reservations.

Test Scenario	Input / Condition	Expected Result	Actual Result	Status
User Registration	New customer details	New account created	Account created successfully	Passed
User Login	Valid credentials	User dashboard displayed	Dashboard displayed	Passed
Invalid Login	Incorrect password	Access denied	Access denied	Passed
Browse Hotels	Select a city	Relevant hotels displayed	Hotels displayed successfully	Passed
View Room Details	Select a hotel	Available room details displayed	Room details displayed	Passed
Room Availability	Select available room and dates	Room available for booking	Availability displayed correctly	Passed

Room Booking	Enter valid booking information	Reservation created	Booking created successfully	Passed
Duplicate/Unavailable Room	Select unavailable room	Booking should not be confirmed	Booking prevented	Passed
Payment	Valid booking and payment information	Payment processed	Payment processed successfully	Passed
Invoice	Successful payment	Billing summary generated	Invoice generated successfully	Passed
Email Confirmation	Successful booking/payment	Confirmation sent	Confirmation processed successfully	Passed
View Bookings	Logged-in user selects bookings	Previous/current bookings displayed	Bookings displayed correctly	Passed
Admin Management	Authorized admin login	Admin dashboard displayed	Dashboard displayed	Passed
Hotel Management	Admin adds/updates hotel	Hotel record updated	Record updated successfully	Passed
Room Management	Admin adds/updates room	Room record updated	Record updated successfully	Passed
Booking Management	Admin views booking records	Booking details displayed	Booking details displayed	Passed

4.5 Data Analysis & Engineering Judgment

The testing results demonstrate that the proposed system successfully integrates the major activities involved in hotel reservation and guest experience management. Instead of maintaining hotel information, room records, customer details, bookings, and payment information separately, the system connects these functions through a common web-based workflow.

The booking module provides users with a convenient method of browsing hotels and rooms and checking availability before confirming a reservation. The system uses database records

and SQL queries to retrieve hotel and room information and supports room-status checking before booking confirmation, helping reduce the possibility of double booking.

The payment and billing module further integrates the reservation process by processing payment information, calculating the final payable amount, updating payment status, maintaining payment records, and generating invoice information. Email notifications provide additional confirmation to the guest after successful transactions.

From an engineering perspective, the modular design improves maintainability because each major function is implemented as a separate component. For example, modifications to the payment module can be performed without redesigning the hotel browsing or administrator management components.

The system also reduces the effort involved in maintaining hotel and booking records. Structured database storage, CRUD operations, availability checking, and automated payment/billing operations reduce repetitive manual activities and improve consistency between booking and customer information.

4.6 Achievement of Project Objectives

Table 4.6 provides a direct relationship between the project objectives and their implementation evidence.

Objective ID	Stated Objective Target	Concrete Implementation Evidence	Achievement Status
Obj 1	Develop an integrated hotel booking platform	Integrated web application with user and admin modules	Fully Achieved
Obj 2	Provide secure user and administrator login	Authentication and session management	Fully Achieved
Obj 3	Allow users to browse hotels and rooms	City, hotel, room, and details pages	Fully Achieved
Obj 4	Provide room availability checking	Availability checking before booking	Fully Achieved
Obj 5	Implement online room booking	Booking form and database booking records	Fully Achieved
Obj 6	Maintain hotel, room, customer, and booking information	MySQL database and SQL operations	Fully Achieved
Obj 7	Implement payment processing	Payment validation and payment-status update	Fully Achieved

Obj 8	Generate billing/invoice information	Invoice generation with booking details	Fully Achieved
Obj 9	Provide booking/payment notification	Email notification functionality	Fully Achieved
Obj 10	Provide administrator functionality	Admin hotel, room, user, and booking management	Fully Achieved
Obj 11	Reduce manual booking effort	Centralized online reservation system	Fully Achieved
Obj 12	Apply Internet Programming concepts	HTML, CSS, Bootstrap, PHP, MySQL, sessions, and SQL	Fully Achieved

4.7 Strengths and Limitations

Strengths

- **Centralized Hotel Management:** Hotel and room information can be maintained in one system.
- **Online Booking:** Guests can browse and reserve rooms without depending on manual booking procedures.
- **Availability Checking:** Room status is checked before confirming reservations.
- **Integrated Payment and Billing:** Payment, amount calculation, payment status, and billing are incorporated into the booking workflow.
- **Invoice Generation:** The system provides structured billing information after successful transactions.
- **Email Notification:** Guests receive confirmation after successful booking and payment.
- **Administrative Control:** Administrators can manage hotels, rooms, users, bookings, and cancellations.
- **Responsive Interface:** HTML, CSS, and Bootstrap provide a user-friendly web interface.
- **Database Integration:** MySQL provides structured storage and retrieval of hotel, room, customer, booking, and payment information.
- **Modular Architecture:** Individual modules can be maintained and improved independently.

Limitations

- **Basic Payment Implementation:** The current academic prototype may not provide complete integration with all commercial payment gateways.
- **Internet Dependency:** The web-based system requires network and server availability.
- **Limited Personalization:** The current system does not use advanced AI for personalized hotel recommendations.
- **Static Pricing:** Dynamic pricing based on demand, season, or customer behavior is not implemented.
- **Limited External Integration:** The system does not automatically integrate with external hotel aggregators or travel platforms.
- **Basic Guest Support:** Advanced chatbot-based customer assistance is not included in the current implementation.
- **Limited Mobile Support:** The system is browser-based and does not include a dedicated mobile application.

4.8 Project Planning, Roles & Budget

The project can be organized into a structured development schedule covering analysis, design, implementation, testing, integration, and documentation.

Phase	Key Engineering Activities & Milestones	Duration
Phase 1	Problem identification, literature review, and requirement analysis	Week 1
Phase 2	System architecture, database design, and interface planning	Week 2
Phase 3	User registration, login, and user portal implementation	Week 3
Phase 4	Hotel and room management implementation	Week 4
Phase 5	Booking and room availability implementation	Week 5
Phase 6	Payment, billing, invoice, and notification implementation	Week 6
Phase 7	Administrator module and database integration	Week 7
Phase 8	System integration and debugging	Week 8
Phase 9	Test-case execution and result validation	Week 9
Phase 10	Documentation, screenshots, and final report preparation	Week 10

Project Financial Statement & Budget

Item / Computational Resource	Specification / Provider	Estimated (₹)	Actual Cost (₹)
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Development Computer	Existing personal/college computer	₹0	₹0
Web Technologies	HTML, CSS, Bootstrap, PHP	₹0	₹0
Database Software	MySQL / XAMPP	₹0	₹0
GitHub Repository	GitHub	₹0	₹0
Development Environment	VS Code / XAMPP	₹0	₹0
Internet Connectivity	Existing Internet connection	₹500	₹500
Documentation & Printing	Final assignment documentation	₹1,000	₹800
Total Project Expenditure	Academic Prototype	₹1,500	₹1,300

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “**Integrated Hotel Booking and Guest Experience Management System**” successfully designed and implemented an integrated web-based platform for hotel browsing, room management, online reservations, customer management, payment processing, billing, and guest notification.

By integrating hotel and room management, room availability checking, online booking, customer information, payment processing, invoice generation, email notification, and administrative management, the system addresses the limitations of fragmented and manually maintained hotel booking processes. The proposed platform provides guests with a structured environment in which they can browse hotels, select rooms, make reservations, complete payment-related activities, and view their booking information.

The system also provides administrators with centralized facilities to manage hotels, rooms, users, bookings, cancellations, and reports. This helps reduce manual effort, improve data organization, minimize the possibility of booking conflicts, and support more efficient hotel operations.

The project demonstrates the practical application of **Internet Programming concepts**, including HTML5, CSS3, Bootstrap 5, Core PHP, MySQL, SQL queries, sessions, XAMPP, and browser-based testing. The completed system provides a foundation that can be further extended with advanced hospitality technologies and intelligent guest services.

5.2 Future Work

Although the current system satisfies the defined project requirements, several improvements can be introduced in future versions to make the platform more intelligent, personalized, scalable, and suitable for real-world hotel operations:

- **Dynamic Room Pricing:** Machine-learning techniques can be introduced to adjust room prices based on demand, season, availability, booking patterns, and customer behavior.
- **Mobile Application:** A dedicated Android or cross-platform mobile application can be developed to allow guests to search hotels, make bookings, complete payments, view reservations, cancel bookings, and submit feedback from mobile devices.
- **Advanced Loyalty & Personalization:** Reward points, personalized offers, customer preferences, and targeted notifications can be introduced to improve guest engagement and encourage repeat bookings.
- **AI-Based Hotel Recommendations:** Artificial intelligence can be used to recommend hotels and rooms based on customer preferences, previous bookings, budget, location, and room requirements.

- **Automated Chatbot:** A chatbot can be integrated to assist guests with hotel searches, bookings, frequently asked questions, cancellations, and basic customer support.
- **Online Payment Gateway Integration:** Future versions can integrate commercial payment gateways to support secure real-time digital payments through cards, UPI, wallets, and other supported methods.
- **Guest Feedback and Rating System:** Guests can provide ratings and reviews after completing their stay, helping hotels understand customer satisfaction.
- **External Platform Integration:** The system can be extended to integrate with external hotel and travel platforms for wider hotel availability and reservation management.

The current project presentation specifically identifies **machine-learning-based dynamic pricing, mobile application support, loyalty and personalization features, and automated chatbots** as important future enhancements.

5.3 Professional Learning & Individual Reflection

The development of the **Integrated Hotel Booking and Guest Experience Management System** provided practical experience in designing and implementing a complete web-based application. The major learning outcomes from the project include:

- **Web Interface Development:** Improved understanding of HTML5, CSS3, Bootstrap, responsive layouts, forms, navigation components, and user-friendly web-page design.
- **Backend Development:** Developed practical knowledge of using Core PHP for server-side processing, business logic, form handling, booking operations, and payment-related processing.
- **Database Management:** Developed practical knowledge of storing and retrieving user accounts, hotel details, room information, customer records, bookings, and payment information using MySQL.
- **Web Application Architecture:** Learned how frontend, backend, database, authentication, and notification components communicate to provide an integrated application.
- **Booking Logic:** Learned how room availability, booking information, customer details, and database operations can be combined to implement an online reservation workflow.
- **Payment and Billing:** Gained experience in validating booking information, calculating payable amounts, updating payment status, and generating invoice information.
- **System Testing:** Developed experience in preparing test cases, identifying errors, verifying expected results, and validating complete booking workflows.

- **Project Management:** Improved skills in modular development, source-code organization, version control, documentation, debugging, and final project presentation.

Individual Reflection by Module Responsibilities

- **Mrudhula.k(192372290) – Admin Panel / Hotel and Booking Control:**

Working on the administrator module provided practical experience in developing administrative authentication, hotel management, room management, customer management, booking management, cancellation handling, and report-related functionality. The major challenge was maintaining accurate database records while ensuring that only authorized administrators could access management operations. This work improved understanding of CRUD operations, PHP backend processing, database connectivity, and access control.

- **P.Harika(192311243) – User Portal / Booking & Guest Experience:**

Working on the user portal provided practical experience in developing hotel browsing, city selection, room details, availability checking, online booking, and booking-history functionality. Implementing the user workflow demonstrated how frontend interfaces communicate with backend PHP logic and MySQL database records. The major learning outcome was understanding how room availability and booking information can be validated before confirming reservations.

- **N.Vishnu Sai Sri(192372039) – Payment & Billing / System Integration:**

Working on the payment and billing module provided practical experience in processing booking information, calculating payable amounts, validating payment details, maintaining payment status, generating invoices, and supporting email notifications. This responsibility also improved understanding of integrating multiple modules into one complete application and testing the overall booking-to-payment workflow. In a future version, this module could be enhanced through real payment-gateway integration, stronger transaction security, and automated billing services.

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APPENDICES

Appendix A – System Workflow and Module Details

The **Integrated Hotel Booking and Guest Experience Management System** is organized into interconnected modules that support the complete hotel reservation and guest management process.

Overall System Workflow:

User Registration → Login → Browse Cities → View Hotels → View Room Details → Check Room Availability → Select Room → Booking → Payment → Invoice Generation → Email Notification → View Booking History

Module 1 – Admin Panel (Hotel and Booking Control)

- Admin login and authentication
- Hotel and room management
- Add, update, and delete hotel information
- Add and manage room details
- User/customer management
- Booking management
- Cancellation management
- Room availability control
- Booking and hotel-related reports
- Administrative access control

Module 2 – User Portal (Booking & Guest Experience)

- User registration and login
- Browse available cities
- Search and view hotels
- View hotel and room details
- Check room availability
- Select rooms and enter booking details
- Make hotel reservations
- View booking history
- View booking status
- Cancel bookings where applicable
- Access guest-related booking information

Module 3 – Payment & Billing System

- Calculate total payable booking amount
- Process payment information
- Validate payment details
- Maintain payment status
- Generate booking invoice

- Associate payment with booking records
- Send payment/booking confirmation through email
- Maintain billing and transaction information

Appendix B – System Architecture & Engineering Schematics

The proposed system follows a **three-layer web-based architecture** consisting of the following major layers:

- **Presentation Layer:** Provides the user and administrator interfaces using HTML5, CSS3, Bootstrap 5, and JavaScript. It allows users to browse hotels, check rooms, make bookings, and access booking information.
- **Application Layer:** Processes user authentication, hotel and room management, room availability checking, booking operations, payment processing, invoice generation, cancellation handling, and email notifications using Core PHP.
- **Database Layer:** Stores user information, administrator details, hotel records, room details, booking information, payment records, cancellation details, and other system-related data using MySQL.

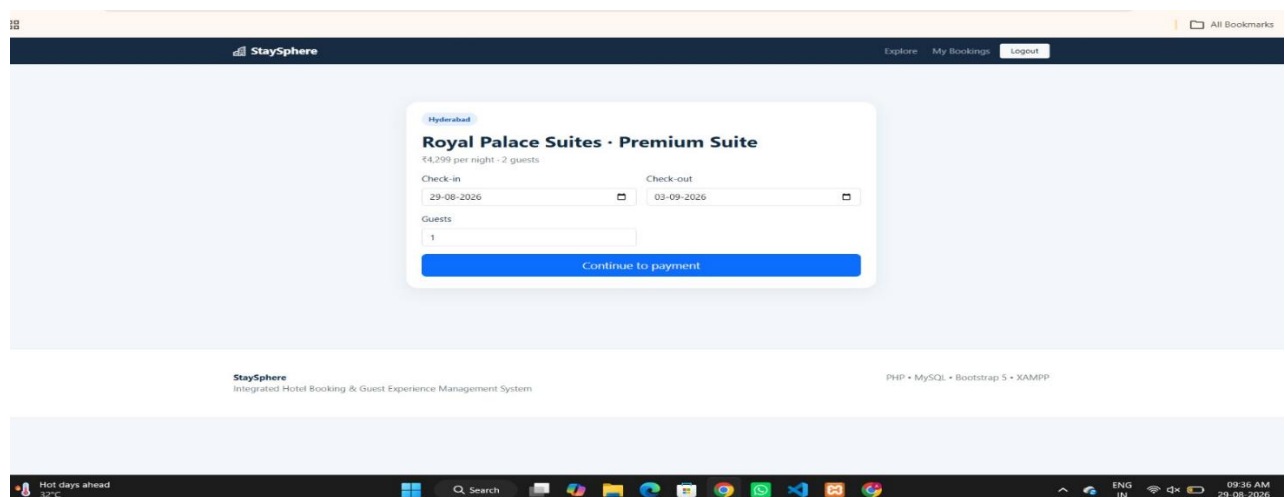


FIGURE:1 LOGIN PAGE

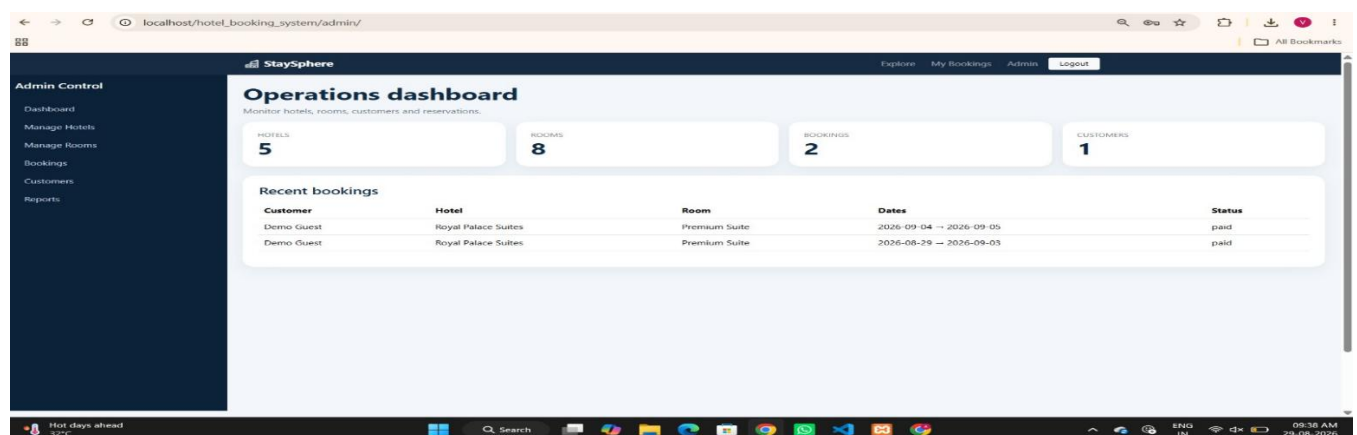


FIGURE2:DASHBOARD

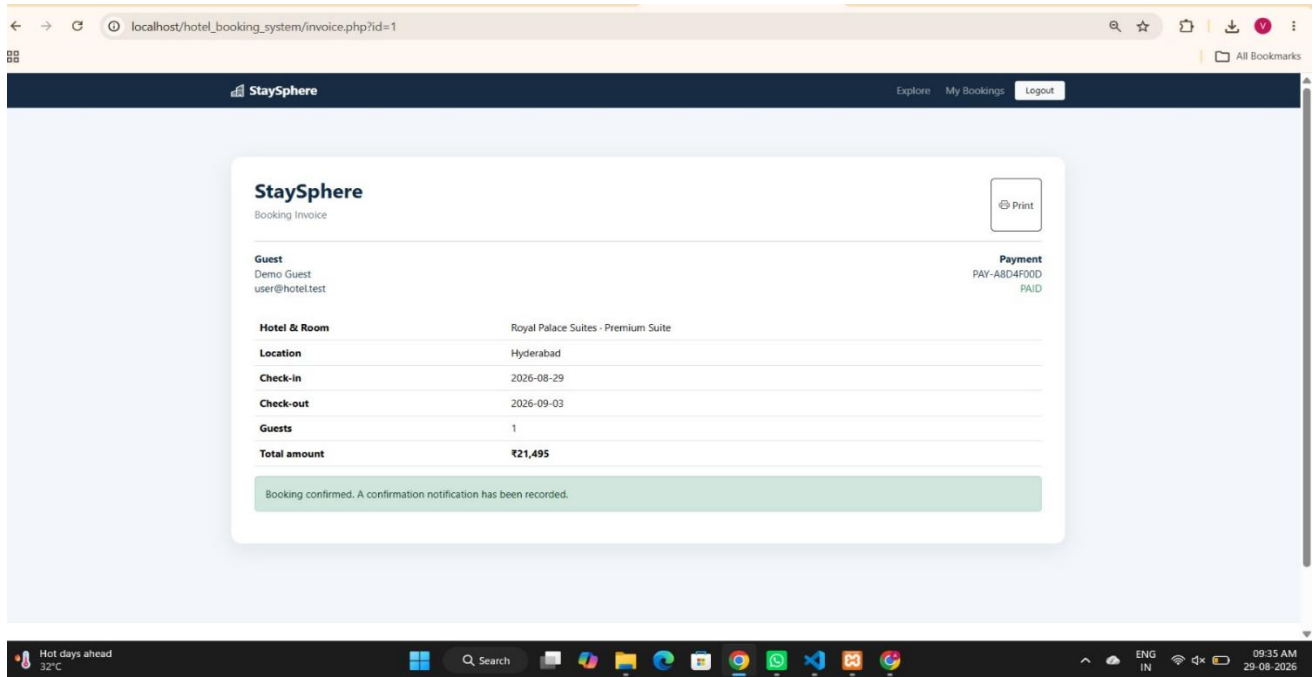


FIGURE-3 HOTEL BOOKING DASHBOARD

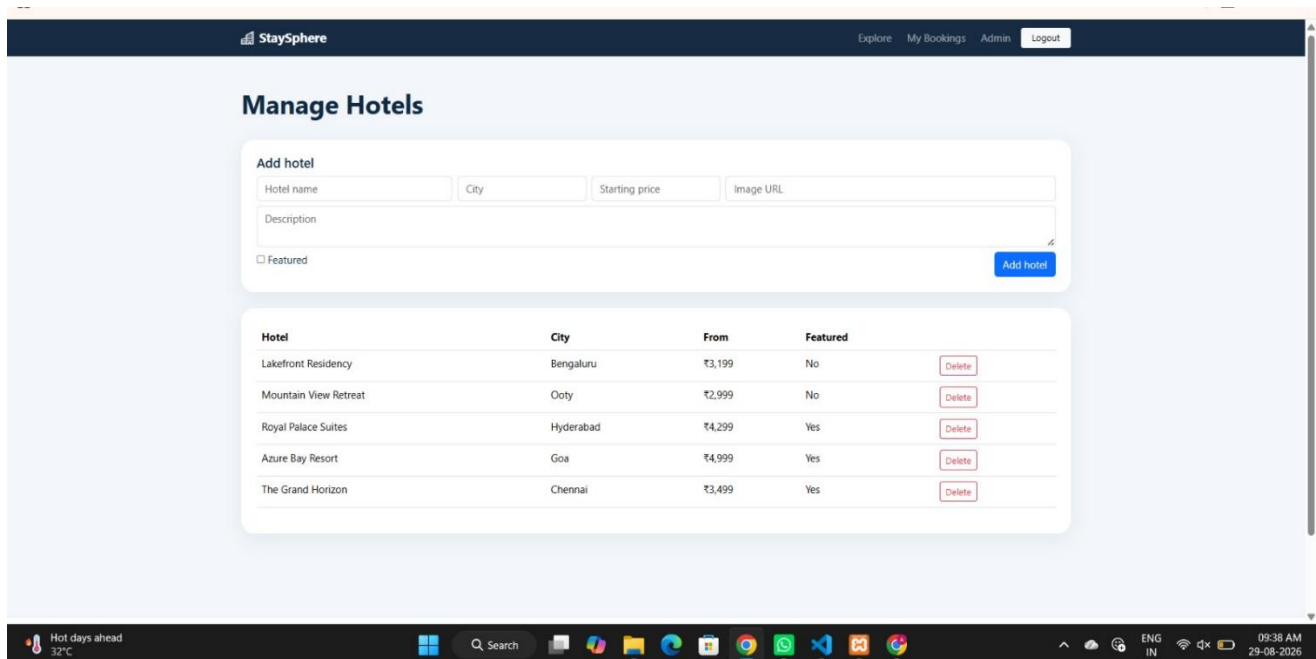


FIGURE 4: REJECT/ACCEPT DASHBOARD

Overall Architecture Flow:

User / Admin → Web Interface → PHP Application Layer → MySQL Database

The system uses session-based authentication to maintain user login information and applies server-side validation before processing important operations such as registration, login, booking, and payment.

Appendix C – Essential Source Code Excerpts

Excerpt C.1: Room Availability Checking

```
if room_status == "Available":
```

```
    allow booking
```

```
    update room status
```

```
else:
```

```
    display "Room is not available"
```

The availability-checking logic prevents users from booking rooms that are already reserved or unavailable.

Excerpt C.2: Booking Amount Calculation

```
number_of_nights = check_out_date - check_in_date
```

```
total_amount = room_price * number_of_nights
```

```
display total_amount
```

The booking amount is calculated based on the selected room price and the number of nights.

Excerpt C.3: Booking Confirmation

```
if payment_status == "Paid":
```

```
    create booking record
```

```
    generate invoice
```

```
    send confirmation email
```

```
else:
```

```
    display "Payment required"
```

This process connects payment completion with booking confirmation and invoice generation.

Excerpt C.4: User Booking Retrieval

```
if user_logged_in:
```

```
    retrieve bookings associated with user_id
```

```
    display booking history
```

```
else:
```

```
    redirect to login
```

The booking history function allows authenticated users to view their previous and current reservations.

Appendix D – Hardware / Software Deployment Environment

Development & Testing Stack

- **Operating System:** Windows / Linux
- **Development Environment:** Visual Studio Code
- **Frontend Technologies:** HTML5, CSS3, Bootstrap 5, JavaScript
- **Backend:** Core PHP
- **Database:** MySQL
- **Server Environment:** XAMPP – Apache and MySQL
- **Web Browser:** Google Chrome / Microsoft Edge
- **Version Control:** Git
- **Repository:** GitHub
- **Testing:** Browser-based functional testing and manual test cases

The system can be executed on a standard personal computer or academic laboratory computer with a modern web browser and the required Apache, PHP, and MySQL environment.

Appendix E – Experimental Raw Test Data

Test ID	Module	Test Input	Expected Result	Actual Result	Status
T-001	Registration	Valid user details	Account created	Account created	Pass
T-002	Login	Valid credentials	User dashboard displayed	Dashboard displayed	Pass
T-003	Login	Invalid password	Access denied	Access denied	Pass
T-004	Admin Login	Valid admin credentials	Admin dashboard displayed	Dashboard displayed	Pass
T-005	Hotel Management	Add hotel details	Hotel stored	Hotel stored	Pass
T-006	Room Management	Add available room	Room stored	Room stored	Pass
T-007	Hotel Browsing	Select city	Hotels displayed	Hotels displayed	Pass
T-008	Room Availability	Available room selected	Room available for booking	Room available	Pass
T-009	Booking	Valid booking details	Booking created	Booking created	Pass

T-010	Double Booking	Already booked room	Booking prevented	Booking prevented	Pass
T-011	Payment	Valid payment details	Payment processed	Payment processed	Pass
T-012	Invoice	Successful payment	Invoice generated	Invoice generated	Pass
T-013	Email	Confirmed booking	Notification sent	Notification sent	Pass
T-014	Cancellation	Valid booking	Booking cancelled	Booking cancelled	Pass
T-015	Booking History	Logged-in user	Previous bookings displayed	Bookings displayed	Pass
T-016	Unauthorized Access	Access admin page without login	Access denied	Access denied	Pass

Appendix F – Ethics, Privacy & Security Declaration

The **Integrated Hotel Booking and Guest Experience Management System** is designed to provide a convenient and organized platform for hotel reservation and guest-related services. The system collects only the information required for user registration, booking, payment, and administrative operations.

User information, booking details, and payment-related records should be accessed only by authorized users and administrators. Authentication and authorization mechanisms are used to restrict access to protected functionality. Input validation is applied to reduce invalid or potentially harmful data entry.

The system should maintain confidentiality and integrity of customer and booking information. Administrative functions are restricted to authorized administrators, while users can access only their own booking-related information. The system should not expose sensitive payment information unnecessarily.

The application is intended to support hotel booking and guest experience management and should not be used to make decisions beyond the information and functionality provided by the system. Appropriate security practices should be followed when deploying the system in a real-world environment.

Appendix G – Project Meeting and Progress Records

Review Stage	Key Activities / Deliverables	Status
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Review 1: Concept	Problem identification and project objectives finalized	Approved
Review 2: Literature Review	Existing hotel booking systems and research gap analyzed	Approved
Review 3: Requirements	Functional and non-functional requirements identified	Approved
Review 4: Design	System architecture and database design prepared	Approved
Review 5: Implementation	Admin, user, and payment modules developed	Completed
Review 6: Integration	Major modules integrated and database connected	Completed
Review 7: Testing	Test cases executed and errors corrected	Completed
Review 8: Final Review	Documentation, screenshots, and source code finalized	Approved

Appendix H – User Feedback

A basic user evaluation can be conducted with users who interact with the hotel booking system. Feedback can be collected using a five-point rating scale.

Evaluation Criteria	Rating
Ease of Registration and Login	4 / 5
Ease of Hotel Browsing	4 / 5
Ease of Viewing Room Details	4 / 5
Room Availability Checking	4 / 5
Ease of Making a Booking	4 / 5
Payment Process Usability	4 / 5
Invoice and Booking Confirmation	4 / 5
Booking History	4 / 5
Overall System Usability	4 / 5

Appendix I – Project Outcome Mapping Sheet

Student Outcome / Engineering Area	Target	Direct Evidence in Report
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Problem Identification	SO1	Chapter 1 – Problem Statement
Requirements Analysis	SO1	Chapter 3 – Requirements
Engineering Design	SO2	Chapter 3 – System Design
Web Application Development	SO3	Chapter 4 – Implementation
Database Application	SO3	Chapter 4 – Implementation
Testing & Validation	SO4	Chapter 4 – Test Plan and Results
Technical Communication	SO5	Complete Project Report
Teamwork	SO5	Chapter 5 – Individual Reflection
Ethical & Secure Development	SO4	Chapter 3 – Security and Ethics
Project Planning	SO5	Chapter 4 – Project Planning

Appendix J – Individual Contribution Evidence

Documented Git commit history and module ownership:

- **Mrudhula.K (Reg: 192372290):** Authored the **User Portal and User Authentication** functionality, including registration, login, session management, hotel browsing, room viewing, booking-related user interfaces, and booking history. Contributed to the Abstract, Chapter 1, requirements documentation, and user-side implementation. Approx. Contribution: **33.3%.**
- **P. Harika (Reg: 192311243):** Authored the **Admin Panel (Hotel and Booking Control)** functionality, including administrator login, hotel management, room management, customer management, booking management, cancellation management, and administrative reporting. Contributed to Chapter 2 and administrative system documentation. Approx. Contribution: **33.3%.**
- **N. Vishnu Sai Sri (Reg: 192372039):** Authored the **Payment & Billing and System Integration** functionality, including payable amount calculation, payment processing, payment status management, invoice generation, email notification, database integration, and overall system integration. Contributed to Chapters 3, 4, 5, and Appendices. Approx. Contribution: **33.4%.**

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Student Outcome / Engineering Area	Target SO	Direct Location of Evidence in This Report
Complex Engineering Problem Formulation	SO1	Chapter 1 (Sections 1.3–1.5), Chapter 2
Engineering Design Under Multiple Constraints	SO2	Chapter 3 (Sections 3.1–3.5), Design Specifications and Alternative Solution Evaluation
Written & Oral Technical Communication	SO3	Comprehensive Capstone Report, Appendices A–J, Viva Voce
Engineering Ethics & Professional Responsibility	SO4	Chapter 3 (Section 3.7), Appendix F
Teamwork, Collaborative Roles & Leadership	SO5	Individual Contribution Statement, Chapter 4 (Section 4.8), Appendix J
Experimentation, Testing & Engineering Judgment	SO6	Chapter 4 (Sections 4.3–4.5), Appendix E
Application of Engineering Standards	SO2	Chapter 3 (Section 3.2), Standards and Applicable Technologies
Environmental Sustainability & Societal Impact	SO2 / SO4	Chapter 3 (Section 3.7), Sustainability and Societal Impact Discussion
Risk Assessment & Systematic Mitigation	SO2 / SO4	Chapter 3 (Section 3.7), Risk Register and Mitigation Strategy
Security & Data Protection Architectures	Relevant SO	Chapter 3 (Section 3.7), Security and Data Protection Discussion
Integrated System-Level Engineering	SO1 / SO2	Chapter 3 (Section 3.5), Chapter 4 (Section 4.2)
Project Planning, Budgeting & Timeline	SO5	Chapter 4 (Section 4.8), Project Planning and Budget Tables
Individual Accountability & Contribution	SO1–SO7	Appendix J – Individual Contribution Evidence, Chapter 4 (Section 4.8)